GLOBAL GENERATION OF AN INVERTIBLE SHEAF OF DEGREE AT LEAST 2H^1 ON EVERY REDUCED CONNECTED CLOSED SUBSCHEME OF DIMENSION 1 OF A PROPER CURVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the author's own induction with the purely transcendental field replacement, the ideal-sheaf contradiction setup, the duality dimension count, the generic-point subspace argument, the factorization through Y with both commutative diagrams, and the closing boundary-map contradiction are all retained verbatim.

Editorial note. The label lemma-global-generation names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Duality for a proper curve. Same chapter, Serre duality on a proper scheme of dimension 1 over a field: H^1(X, F) is dual to Hom(F, omega_X), used with Lemma lemma-vanishing-on-gorenstein and the deg-chi formula. Supplies the dimension identity dim H^0(X, IL) = deg(L) + chi(O_X) > dim H^0(X, omega_X) that forces the chosen map phi to vanish on global sections.
- Behaviour of global generation and genus under base field extension. Varieties, Lemmas varieties-lemma-globally-generated-base-change, varieties-lemma-degree-base-change, varieties-lemma-geometrically-reduced-any-base-change, varieties-lemma-bijection-irreducible-components; Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology. Justifies replacing the ground field k by a purely transcendental extension K, so the union-of-subspaces argument may assume k infinite.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Prerequisite pointer invoked at proof line(s) 5780; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E33: sources/stacks-curves/pinned-original-source.tex, source lines 976–976. Prerequisite pointer invoked at proof line(s) 5849; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0E3C: sources/stacks-curves/pinned-original-source.tex, source lines 5593–5593. Prerequisite pointer invoked at proof line(s) 5770; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E32: sources/stacks-curves/pinned-original-source.tex, source lines 927–927. Prerequisite pointer invoked at proof line(s) 5781; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3E: sources/stacks-curves/pinned-original-source.tex, source lines 5680–5680. Prerequisite pointer invoked at proof line(s) 5811; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 038F: sources/stacks-chow/tags.txt, source lines 4234–4234. Prerequisite pointer invoked at proof line(s) 5779; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Prerequisite pointer invoked at proof line(s) 5801; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B59: sources/stacks-chow/tags.txt, source lines 14027–14027. Prerequisite pointer invoked at proof line(s) 5777; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 035Z: sources/stacks-chow/tags.txt, source lines 4146–4146. Prerequisite pointer invoked at proof line(s) 5778; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B57: sources/stacks-chow/tags.txt, source lines 14025–14025. Prerequisite pointer invoked at proof line(s) 5776; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k which is reduced, connected, and of dimension 1. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Assume that for every reduced connected closed subscheme $Y \subset X$ of dimension 1 we have*

$$\deg(\mathcal{L}|_Y) \geq 2 \dim_k H^1(Y, \mathcal{O}_Y)$$

Then $\mathcal{L}$ is globally generated.

Proof. By induction on the number of irreducible components of X. If X is irreducible, then the lemma holds by Lemma 0E3C applied to X viewed as a scheme over the field $k' = H^0(X, \mathcal{O}_X)$. Assume X is not irreducible. Before we continue, if k is finite, then we replace k by a purely transcendental extension K. This is allowed by Varieties, Lemmas 0B57, 0B59, 035Z, and 038F, Cohomology of Schemes, Lemma 02KH, Lemma 0E32 and the elementary fact that K is geometrically integral over k.

Assume that $\mathcal{L}$ is not globally generated to get a contradiction. Then we may choose a coherent ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$ such that $H^0(X, \mathcal{I}\mathcal{L}) = H^0(X, \mathcal{L})$ and such that $\mathcal{O}_X/\mathcal{I}$ is nonzero with support of dimension 0. For example, take $\mathcal{I}$ the ideal sheaf of any closed point in the common vanishing locus of the global sections of $\mathcal{L}$. We consider the short exact sequence

$$0 \to \mathcal{I}\mathcal{L} \to \mathcal{L} \to \mathcal{L}/\mathcal{I}\mathcal{L} \to 0$$

Since the support of $\mathcal{L}/\mathcal{I}\mathcal{L}$ has dimension 0 we see that $\mathcal{L}/\mathcal{I}\mathcal{L}$ is generated by global sections (Varieties, Lemma 0AYT). From the short exact sequence, and the fact that $H^0(X, \mathcal{I}\mathcal{L}) = H^0(X, \mathcal{L})$ we get an injection $H^0(X, \mathcal{L}/\mathcal{I}\mathcal{L}) \to H^1(X, \mathcal{I}\mathcal{L})$.

Recall that the k-vector space $H^1(X, \mathcal{I}\mathcal{L})$ is dual to $\mathrm{Hom}(\mathcal{I}\mathcal{L}, \omega_X)$. Choose $\varphi : \mathcal{I}\mathcal{L} \to \omega_X$. By Lemma 0E3E we have $H^1(X, \mathcal{L}) = 0$. Hence

$$\dim_k H^0(X, \mathcal{I}\mathcal{L}) = \dim_k H^0(X, \mathcal{L}) = \deg(\mathcal{L}) + \chi(\mathcal{O}_X) > \dim_k H^1(X, \mathcal{O}_X) = \dim_k H^0(X, \omega_X)$$

We conclude that φ is not injective on global sections, in particular φ is not injective. For every generic point $\eta \in X$ of an irreducible component of X denote $V_\eta \subset \mathrm{Hom}(\mathcal{I}\mathcal{L}, \omega_X)$ the k-subvector space consisting of those φ which are zero at η. Since every associated point of $\mathcal{I}\mathcal{L}$ is a generic point of X, the above shows that $\mathrm{Hom}(\mathcal{I}\mathcal{L}, \omega_X) = \bigcup V_\eta$. As X has finitely many generic points and k is infinite, we conclude $\mathrm{Hom}(\mathcal{I}\mathcal{L}, \omega_X) = V_\eta$ for some η. Let $\eta \in C \subset X$ be the corresponding irreducible component. Let $Y \subset X$ be the union of the other irreducible components of X. Then Y is a nonempty reduced closed subscheme not equal to X. Let $\mathcal{J} \subset \mathcal{O}_X$ be the ideal sheaf of Y. Please keep in mind that the support of $\mathcal{J}$ is C.

Let $\varphi : \mathcal{I}\mathcal{L} \to \omega_X$ be arbitrary. Since $\mathcal{J}\mathcal{I}\mathcal{L}$ has no embedded associated points (as a submodule of $\mathcal{L}$) and as φ is zero in the generic point η of the support of $\mathcal{J}$, we find that φ factors as

$$\mathcal{I}\mathcal{L} \to \mathcal{I}\mathcal{L}/\mathcal{J}\mathcal{I}\mathcal{L} \to \omega_X$$

We can view $\mathcal{I}\mathcal{L}/\mathcal{J}\mathcal{I}\mathcal{L}$ as the pushforward of a coherent sheaf on Y which by abuse of notation we indicate with the same symbol. Since $\omega_Y = \mathcal{H}om(\mathcal{O}_Y, \omega_X)$ by Lemma 0E33 we find a factorization

$$\mathcal{I}\mathcal{L} \to \mathcal{I}\mathcal{L}/\mathcal{J}\mathcal{I}\mathcal{L} \xrightarrow{\varphi'} \omega_Y \to \omega_X$$

of φ. Let $\mathcal{I}' \subset \mathcal{O}_Y$ be the image of $\mathcal{I} \subset \mathcal{O}_X$. There is a surjective map $\mathcal{I}\mathcal{L}/\mathcal{J}\mathcal{I}\mathcal{L} \to \mathcal{I}'\mathcal{L}|_Y$ whose kernel is supported in closed points. Since ω_Y is a Cohen-Macaulay module on Y, the map φ' factors through a map $\varphi'' : \mathcal{I}'\mathcal{L}|_Y \to \omega_Y$. Thus we have commutative diagrams

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{I}\mathcal{L} & \longrightarrow & \mathcal{L} & \longrightarrow & \mathcal{L}/\mathcal{I}\mathcal{L} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{I}'\mathcal{L}|_Y & \longrightarrow & \mathcal{L}|_Y & \longrightarrow & \mathcal{L}|_Y/\mathcal{I}'\mathcal{L}|_Y & \longrightarrow & 0 \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathcal{I}\mathcal{L} & \xrightarrow{\varphi} & \omega_X \\ \downarrow & & \uparrow \\ \mathcal{I}'\mathcal{L}|_Y & \xrightarrow{\varphi''} & \omega_Y \end{array}$$

Now we can finish the proof as follows: Since for every φ we have a φ'' and since $\omega_X \in Coh(\mathcal{O}_X)$ represents the functor $\mathcal{F} \mapsto \mathrm{Hom}_k(H^1(X, \mathcal{F}), k)$, we find that $H^1(X, \mathcal{I}\mathcal{L}) \to H^1(Y, \mathcal{I}'\mathcal{L}|_Y)$ is injective. Since the boundary $H^0(X, \mathcal{L}/\mathcal{I}\mathcal{L}) \to H^1(X, \mathcal{I}\mathcal{L})$ is injective, we conclude the composition

$$H^0(X, \mathcal{L}/\mathcal{I}\mathcal{L}) \to H^0(X, \mathcal{L}|_Y/\mathcal{I}'\mathcal{L}|_Y) \to H^1(X, \mathcal{I}'\mathcal{L}|_Y)$$

is injective. Since $\mathcal{L}/\mathcal{I}\mathcal{L} \to \mathcal{L}|_Y/\mathcal{I}'\mathcal{L}|_Y$ is a surjective map of coherent modules whose supports have dimension 0, we see that the first map $H^0(X, \mathcal{L}/\mathcal{I}\mathcal{L}) \to H^0(X, \mathcal{L}|_Y/\mathcal{I}'\mathcal{L}|_Y)$ is surjective (and hence bijective). But by induction we have

that $\mathcal{L}|_Y$ is globally generated (if Y is disconnected this still works of course) and hence the boundary map

$$H^0(X, \mathcal{L}|_Y/\mathcal{I}'\mathcal{L}|_Y) \to H^1(X, \mathcal{I}'\mathcal{L}|_Y)$$

cannot be injective. This contradiction finishes the proof. $\square$